Predicting Academic Risk in Higher Education

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Abstract
This project addresses the challenge of predicting academic outcomes for students in higher education. Using the Kaggle Playground Series S4E6 dataset, we developed a classification pipeline to predict whether students will Graduate, Drop out, or remain Enrolled. We explored the data through visualization, built a preprocessing pipeline to handle numerical features, and compared three machine learning models: Logistic Regression, Random Forest, and XGBoost.

1. Introduction

1.1 Problem Statement
This project aims to build a predictive model that classifies students into three categories based on their academic trajectory:

- Graduate - Students who successfully complete their program
- Dropout - Students who leave before completion
- Enrolled - Students who are still active in their program

1.2 Dataset
The dataset comes from Kaggle's Playground Series Season 4, Episode 6. It contains anonymized student records with 37 features covering:

- Demographic information (age, gender, marital status)
- Academic background (previous qualifications, admission grades)
- Enrollment details (course, attendance type)
- First and second semester performance (units enrolled, approved, grades)
- Economic indicators (unemployment rate, inflation, GDP)

Dataset	Samples	Features
Training	76,518	37
Test	51,012	37

2. Exploratory Data Analysis

2.1 Data Quality

The dataset was clean with no missing values across all 37 features. This simplified our preprocessing since we did not need extensive imputation strategies.

	id	Marital status	Application mode	Application order	Course	Daytime/evening attendance	Previous qualification	Previous qualification (grade)	Nacionality	Mother's qualification	...	Curricular units 2nd sem (credited)	Curricular units 2nd sem (enrolled)	Curricular units 2nd sem (evaluations)	Curricular units 2nd sem (approved)	Curricular units 2nd sem (grade)	Curricular units 2nd sem (without evaluations)	Unemployment rate	Inflation rate	GDP	Target
0	0	1	1	1	9238	1	1	126.0	1	1	...	0	6	7	6	12.428571	0	11.1	0.6	2.02	Graduate
1	1	1	17	1	9238	1	1	125.0	1	19	...	0	6	9	0	0.000000	0	11.1	0.6	2.02	Dropout
2	2	1	17	2	9254	1	1	137.0	1	3	...	0	6	0	0	0.000000	0	16.2	0.3	-0.92	Dropout
3	3	1	1	3	9500	1	1	131.0	1	19	...	0	8	11	7	12.820000	0	11.1	0.6	2.02	Enrolled
4	4	1	1	2	9500	1	1	132.0	1	19	...	0	7	12	6	12.933333	0	7.6	2.6	0.32	Graduate

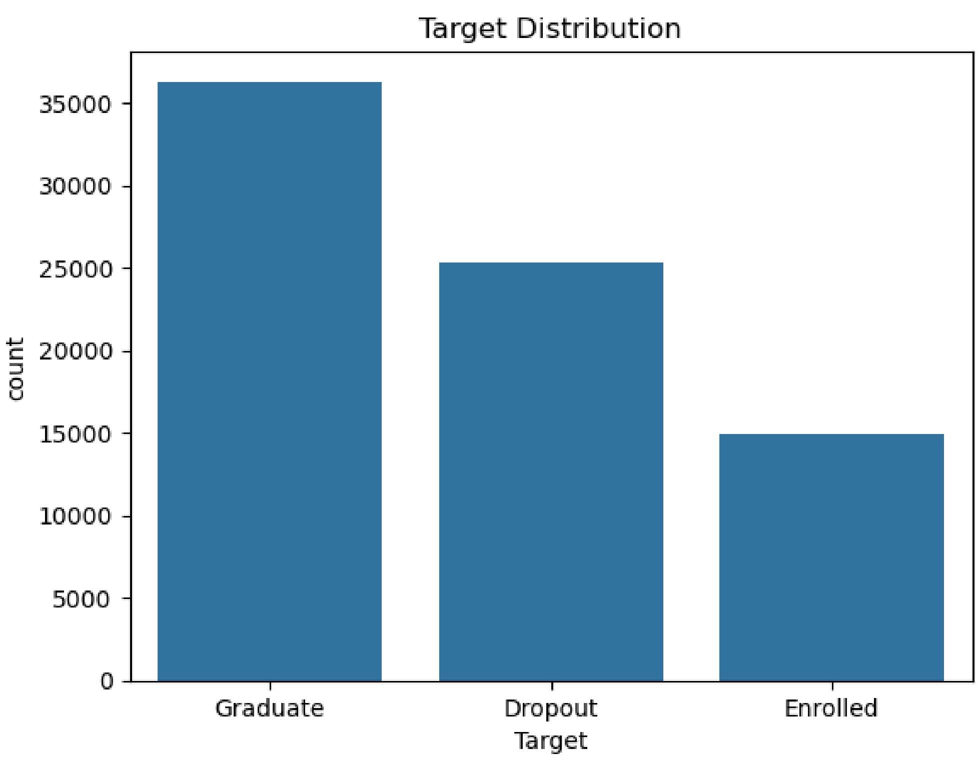
5 rows × 38 columns

2.2 Target Distribution

The target variable shows class imbalance:

Class	Count	Percentage
Graduate	~36,000	47%
Dropout	~25,000	33%
Enrolled	~15,000	20%

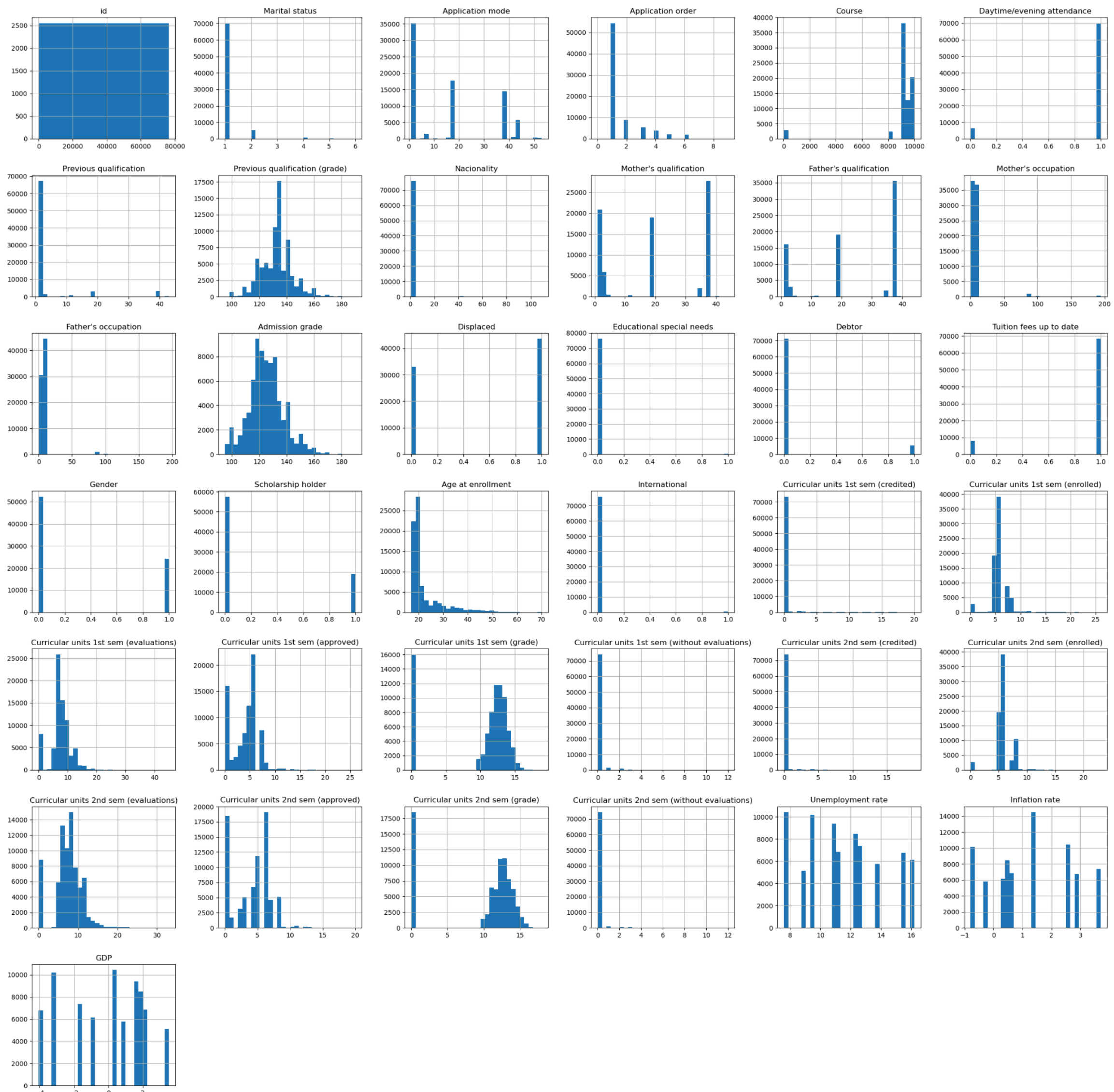
This imbalance informed our decision to use stratified train-test splits and class weights during model training



2.3 Feature Distributions

We visualized all numerical features using histograms to understand their distributions:

- Age at enrollment is right-skewed, with most students between 18-25
- Grade features (admission, qualification) follow roughly normal distributions centered around 120-140
- Binary features (scholarship holder, debtor, international) are heavily skewed toward 0
- Curricular unit features show varied distributions reflecting different student performance levels



3. Methodology

3.1 Preprocessing Pipeline

We built a preprocessing pipeline using scikit-learn's ColumnTransformer to handle different feature types:

Numerical Features:

- Imputation using median values
- Standardization using StandardScaler

Categorical Features:

- Imputation using the most frequent value
- One-hot encoding with unknown category handling

3.2 Train-Validation Split

We used an 80-20 train-validation split with stratification to maintain class proportions in both sets. This gave us 61,214 training samples and 15,304 validation samples.

3.3 Models

We evaluated three classification algorithms:

Logistic Regression (Baseline): Linear model with balanced class weights.

Random Forest: Ensemble of 300 decision trees with balanced class weights.

XGBoost: Gradient boosting with 500 estimators, learning rate of 0.05, max depth of 6.

3.4 Hyperparameter Tuning

We performed RandomizedSearchCV on Random Forest with 3-fold cross-validation:

Parameter	Search Space
n_estimators	[200, 400, 600]
max_depth	[None, 8, 12]
min_samples_split	[2, 5, 10]

Best parameters found: **n_estimators=400, min_samples_split=5, max_depth=None**

4. Results

4.1 Model Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	80.4%	0.922
Random Forest	82.6%	0.931
XGBoost	83.0%	0.936

XGBoost achieved the best performance on both metrics, though all three models performed well with ROC-AUC scores above 0.92.

4.2 Classification Report

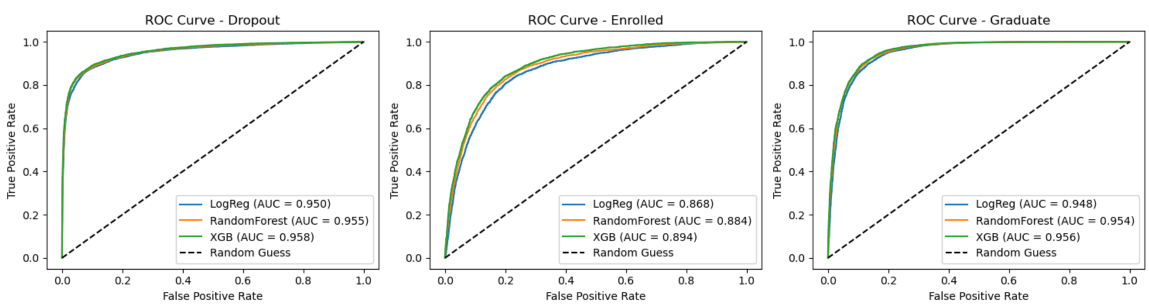
Class	Precision	Recall	F1-Score	Support
Dropout	0.91	0.78	0.84	5,059
Enrolled	0.55	0.72	0.62	2,988
Graduate	0.89	0.85	0.87	7,257

The Enrolled class shows notably lower precision (0.55), indicating this is the most difficult class to predict.

4.3 ROC Curve Analysis

We generated ROC curves for each class using the One-vs-Rest approach:

Class	LogReg AUC	RF AUC	XGB AUC
Dropout	0.950	0.955	0.958
Enrolled	0.868	0.884	0.894
Graduate	0.948	0.954	0.956



The ROC curves confirm that the Enrolled class is harder to predict (AUC ~0.87-0.89) compared to Dropout and Graduate (AUC ~0.95+). This makes sense since enrolled students have not yet completed their academic journey.

4.4 Test Set Predictions

Using our best model (XGBoost), we generated predictions for the 51,012 test samples:

Predicted Class	Count
Graduate	26,647
Dropout	15,258
Enrolled	9,107

The prediction distribution roughly mirrors the training data, indicating that the model has learned meaningful patterns.

5. Discussion

5.1 Key Findings

1. Tree-based models outperform linear models. Both Random Forest and XGBoost achieved higher accuracy and AUC than Logistic Regression, suggesting the presence of non-linear relationships in the data.
2. Enrolled students are the most difficult to classify, as they occupy an intermediate state. Their features may resemble both future graduates and future dropouts.
3. All models achieved an ROC-AUC above 0.92, indicating that the dataset contains useful signals for predicting academic outcomes.

5.2 Limitations

- We did not perform extensive feature engineering or selection
- Only three models were compared

6. Conclusion

We successfully built a machine learning pipeline to predict academic risk for higher education students. Our XGBoost model achieved 83% accuracy and 0.936 ROC-AUC, effectively distinguishing between students who will graduate, drop out, or remain enrolled. The preprocessing pipeline handles both numerical and categorical features, and the stratified cross-validation ensures reliable performance estimates, even in the presence of class imbalance.

6.1 Responsibilities

Arthur Behesnilian	**Lisa Liu**	**Isidoro Flores**	**Osvaldo Valdiviezo**	**Kenia Sanchez-Macario**
Data Analysis, Model Training, Documentation	Documentation, Presentation, Peer Review	Data Analysis, Documentation, Presentation	Model Training, Documentation, Peer Review	Documentation, Presentation, Peer Review

References

- Kaggle Playground Series S4E6: https://www.kaggle.com/competitions/playground-series-s4e6
- scikit-learn documentation: https://scikit-learn.org
- XGBoost documentation: https://xgboost.readthedocs.io